System Health Monitor Overview

Overview & Implementation Steps

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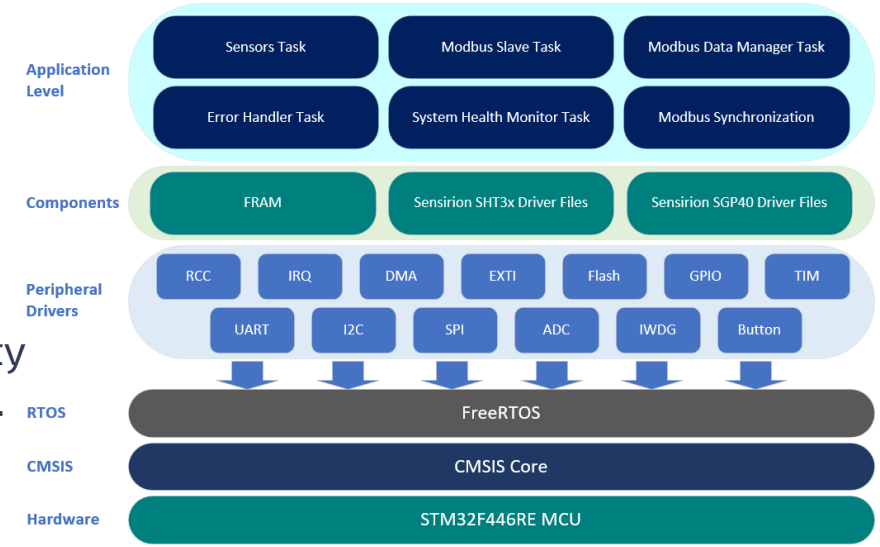
- System Health Monitor Overview
- Section Implementation Steps

System Health Monitor Overview Part I

General

- Purpose & Functionality: Monitors MCU's internal temperature to ensure safe operation and resets the Independent Watchdog Timer (IWDG) if all critical tasks are verified as running.

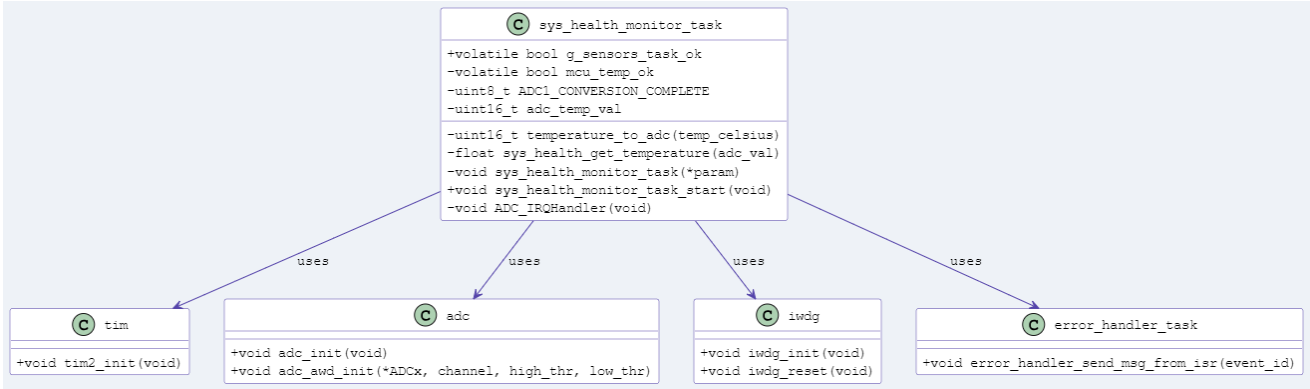
- Project Significance:
  - **Safety Monitoring**: Enhances reliability by monitoring temperature anomalies.
  - **Watchdog Management**: Ensures system responsiveness by verifying that critical tasks are operational.
  - **User Interaction**: Allows users to acknowledge previous watchdog resets upon startup.



System Health Monitor Task Overview Part II

System Health Monitor Task: Monitors MCU's internal temperature, checks critical task status, resets the watchdog timer, and communicates with the error handler.

- **FreeRTOS Task**: Loop with a periodic delay, checks critical tasks and resets IWDG.
- **Temperature Monitoring**: ADC setup on a timer trigger to obtain internal MCU temperature and the ADC's Analog Watchdog (AWD) monitors pre-defined thresholds.
- **Interrupt Handling**: ADC_IRQHandler handles ADC conversion completion, Analog Watchdog monitoring, and communicates with the error handler task.
- Project Significance: The task's design allows for extensibility and serves as a basic example for System Health Monitoring.

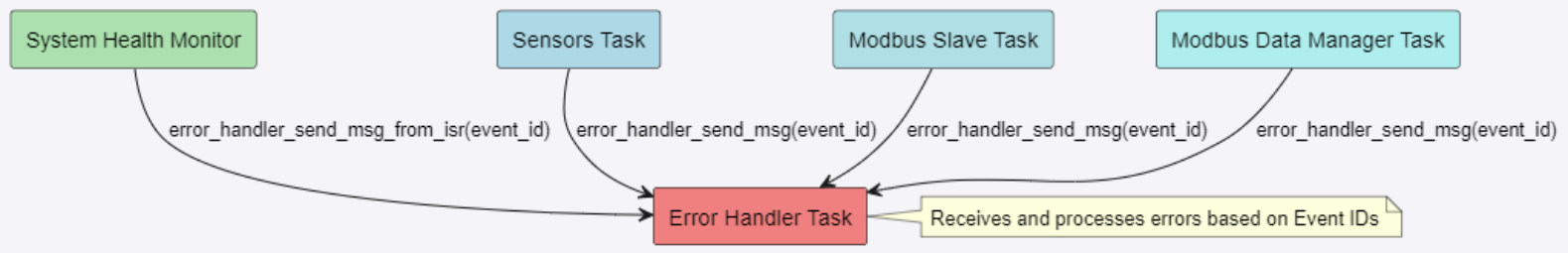


System Health Monitor Task Overview Part III

System Health Monitor Temperature Alarms: Communication facilitated through error_handler_send_msg_from_isr, via the ADC _IRQHandler Analog Watchdog Interrupts.

- **Analog Watchdog**: Monitors converted analog values against high and low thresholds. Triggers an interrupt if values fall outside these limits, detecting abnormal temperature conditions.
- **Communication Mechanism**: ADC_IRQHandler calls error_handler_send_msg_from_isr with EVT_SYS_HEALTH_AWDG_THRESHOLD_EXCEEDED event ID to notify the Error Handler Task.

- Project Significance: Ensures the System Health Monitor has immediate awareness of temperature threshold violations and communicates alarms to the Error Handler.

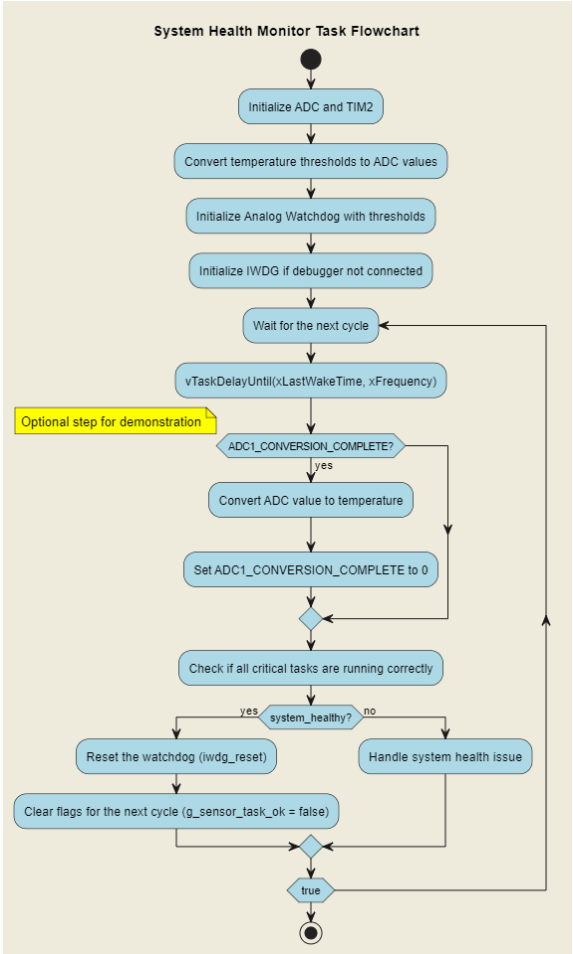


System Health Monitor Task Overview Part IV

Task Overview: Explanation related to the flow diagram.

- **Timer-Triggered ADC**: Transfers the responsibility of getting ADC values and checking the MCU temperature to the ADC_IRQHandler.
- **AWD Initialization**: Allows for customization of the Analog Watchdog's upper and lower thresholds.
- **IWDG Initialization**: Initialized only if the debugger is not connected, ensuring the watchdog is enabled only when necessary.
- **Periodic Critical Task Checks**: Verifies that critical tasks are operational and that the internal MCU temperature is within the acceptable range before resetting the watchdog.

- Project Significance: A basic System Health Monitor for demonstration purposes, which can be expanded for specific application needs.



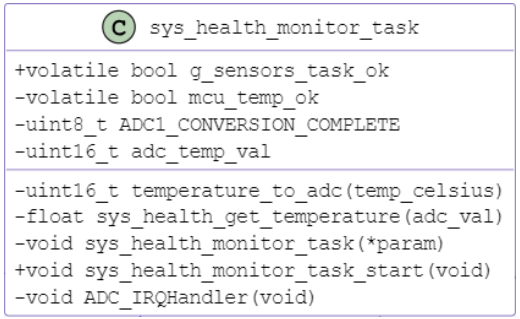
System Health Monitor Files Preparation

Overview

- Objective: Prepare template .c/.h files for integrating additional components as they are developed, e.g., ADC, IRQ, TIM, IWDG, EXTI & Button (to be used at Startup).

- Key Steps:
  - Create sys_health_monitor.h with Analog Watchdog thresholds, critical task flag, and function prototype.
  - Create sys_health_monitor.c with health status flags, ADC conversion completion flag, ADC temperature variable and task functions.

- Project Significance: Sets up the System Health Monitor Task for ongoing development and integration.



ADC, TIM & IRQ Priority Subsection

Overview

- Objective: Provide an overview and implement ADC, TIM, and IRQ drivers required for the System Health Monitor.

- Key Steps:
  - Review the ADC & TIM trigger settings needed for obtaining the internal MCU temperature and IRQ priority requirements for FreeRTOS.
  - Implement ADC initialization (adc.c/.h files).
  - Implement TIM trigger for ADC conversions (tim.c/.h files).
  - Define IRQ priorities and implement the `set priorities` function (irq.c/.h files).

- Project Significance:
  - This development section focuses on achieving the internal MCU temperature monitoring for the System Health Monitor.

System Health Monitor Temperature Implementation & Testing

Overview

- Objective: Integrate IRQ set priorities, ADC & TIM initialization functions, implement the ADC IRQ Handler, Temperature Conversion, and Get Temperature functions.

- Key Steps:
  - **Define Functions**: `temperature_to_adc` and `get_temperature`.
  - **Integration**: Call IRQ Set Priorities, and TIM and ADC Initialization functions.
  - **Define ADC IRQ Handler**: Check if the Analog Watchdog and End of Conversion flags are set and take the appropriate actions.
  - **Testing**: Run the System Health Monitor task and verify ADC interrupts, Internal MCU temperature, and Analog Watchdog interrupts.

- Project Significance: Confirms that the Internal MCU Monitoring portion of the System Health Monitor is operational and prepares for the next steps.

IWDG, EXTI & Button Sub-Section

Overview

- Objective: Provide an overview and implement the IWDG, EXTI, and Button drivers required for the System Health Monitor and user acknowledgment of IWDG resets.

- Key Steps:
  - Review the IWDG configuration, EXTI driver and Button implementation.
  - IWDG implementation (iwdg.c/.h files).
  - EXTI implementation (exti.c/.h files).
  - Button implementation (button.c/.h files).

- Project Significance:
  - Focuses on implementing the Independent Watchdog for the System Health Monitor and user-button acknowledgment for previous IWDG resets at startup.

IWDG/User-Button Acknowledgement Implementation & Testing

Overview

- Objective: Integrate Button initialization and check/acknowledge functions at Startup, and IWDG reset into the System Health Monitor, and perform testing.

- Key Steps:
  - **Integration**: Call `button_init` & `button_check_and_acknowledge_iwdg_event` at Startup and `iwdg_init` & `iwdg_reset` within the System Health Monitor task.
  - **Testing**: Run the project to cause an Independent Watchdog Reset and test USER-button acknowledgement.

- Project Significance: Confirms that the Independent Watchdog portion of the System Health Monitor is operational and that USER-button acknowledgements allow for acknowledging and clearing the event.

Next: System Health Monitor Files Preparation